
 t -distribution

1. Suppose $X_1, \dots, X_5 \stackrel{\text{iid}}{\sim} N(0, 1)$. Find the value c such that $\frac{c(X_1 + X_2)}{\sqrt{X_3^2 + X_4^2 + X_5^2}}$ has a t distribution. What specific t distribution does this random variable have?